(a) Case for Skill-GNN

[1] **Task:** put a cool cup in sidetable → Executor

[2] **Initial Skill Guidance**
object-placer
receptacle-closer
receptacle-preparer
SKILL.md

...

[3] **Failure Feedback**
take cup from countertop
cool cup with fridge → Nothing happens

[4] **Runtime Skill-GNN Query**
goal: cool cup + place in sidetable
required_state: cool
device: fridge
observation: Nothing happens
recent action: failed cooling
GNN

...

[5] **Recovered Guidance + Execution**
retrieve:
location-navigator
receptacle-navigator
Task Complete

Skill-GNN shifts guidance from object placement to prerequisite navigation.

(b) Case for Skill-DAG Template

[1] **Task:** put a cool plate in cabinet → Executor

Reusable successful templates

Template
1. go to diningtable
2. take plate
3. go to fridge
4. open fridge
5. cool plate
6. go to cabinet
7. place plate

Task Complete

Failure Feedback
Invalid attempt:
take plate from diningtable while at cabinet → Nothing happens

Skip!

Compare

Skill-GNN Only
Success: True
Steps: 22
Tokens: 63,340

Exact Skill-DAG Template
Success: True
Steps: 7
Tokens: 13,779
Template ID: traj-442306e1489ae62a

Steps: 22 → 7
Tokens: 63,340 → 13,779

Skill-DAG template reuses subgoal order, turning open-ended search into structured execution.